VOICEOPS 2026: ZENTROMART AI SUPPORT AGENT

Capstone project report | Groq Qwen 3.8 27B + Gnani Vachana

1. What I built

A voice-enabled customer-support assistant for the supplied synthetic ZentroMart dataset. The pipeline is: customer audio -> Gnani STT -> Groq-hosted Qwen -> section-based retrieval and verified tools -> Gnani TTS. The command-line batch runner writes a transcript, reply text, source IDs, and a WAV reply for each input recording.

Step 1 - Prompt baseline. The same support system prompt answers the supplied questions without retrieved company documents. The configured Groq model is **qwen/qwen3.8-27b**.

Step 2 - RAG. Markdown knowledge documents are chunked at stable section IDs. A local sparse TF-IDF-style cosine index includes catalog rows, expands a small synonym list, gives higher priority to policies, and down-ranks superseded documents. Retrieved section IDs are attached to agent replies.

Step 3 - Conversation and tools. The Groq tool-calling agent retains recent conversation turns. The mock **lookup_order** tool requires a matching phone suffix or delivery pincode before exposing order details; **create_ticket** supports human handover. Verification state is limited to the current agent session.

Step 4 - Voice. **GnaniSTTClient** transcribes audio and **GnaniTTSClient** creates replies. The project default is voice Kaveri with **timbre-v2.5**; **GNANI_TTS_MODEL** can override it. Inputs supported by the batch runner: WAV, MP3, M4A, FLAC, and OGG.

2. Step 1 vs Step 2 results

The saved evaluation covers 52 practice questions. Fact scoring checks required phrases and alternatives; retrieval scoring checks whether at least one expected section ID was retrieved. These are automatic string-match metrics, not human ratings.

Measure	Observed result
Prompt-only fact accuracy (49 answerable questions)	10.2%
RAG fact accuracy (49 answerable questions)	44.9%
Retrieved at least one gold section (47 scored cases)	70.2%
Questions triggering forbidden-phrase check	0 / 52
Conversation evaluation	Stopped at PC014: Groq HTTP 429 token limit; no aggregate score

Fact accuracy uses the 49 questions with required facts; retrieval hit rate uses the 47 questions with gold evidence. The conversation suite contains 23 conversations and 96 turns, but its run stopped at PC014 when Groq returned HTTP 429 for the organization's token-per-day limit. No conversation aggregate is reported.

3. What I learned

Grounding improved fact accuracy from 10.2% to 44.9%, a 34.7 percentage-point increase, while the retrieval hit rate was 70.2%. This is useful progress but shows that the current retriever and answer composition still miss many required facts. Zero rows triggered the forbidden-phrase check; that automated check is not a guarantee that every answer is correct.

VOICEOPS 2026: PROJECT REVIEW

Implementation evidence, limitations, and next steps

4. Voice test and tool safety

A sample audio round trip succeeded. The saved test used an .m4a recording asking about an order's delivery. Gnani produced a transcript; the agent asked for the registered phone's last four digits before sharing delivery details; a Gnani WAV reply was saved. This verifies one end-to-end sample, not broad language or noisy-audio coverage.

The system prompt specifies 20 October 2026 as the reference date and Monday-Saturday business days excluding dataset holidays. However, date and business-day calculations are not implemented as a deterministic Python rules engine; answers needing arithmetic still depend on the language model. The order lookup is a synthetic local mock, not production identity verification.

5. What went wrong

The full conversation evaluation could not finish because Groq's on-demand token-per-day limit was reached. The existing local run did not save a final aggregate, so conversation-level accuracy and tool-call accuracy remain unmeasured. The question evaluation also shows room to improve: roughly 3 in 10 scored evidence cases retrieved no gold section, and fewer than half of answerable questions met every required fact.

The scorer is exact phrase based and may penalize correct paraphrases or miss unsupported wording not listed in the forbidden phrases. The single voice sample confirms the pipeline but does not establish latency, robustness, or quality across all supported audio formats and languages.

6. Next improvements

1. Add deterministic Python handling for return deadlines, refund ETAs, and business-day counts using the supplied holiday calendar. 2. Tune retrieval from the missed evidence cases and evaluate ASR-transcribed versions separately. 3. Add per-conversation checkpoints/resume to avoid repeating API work after rate limits. 4. Complete the conversation suite after quota reset and report its aggregate scores. 5. Test Hindi and other supported language codes, longer/noisy recordings, and playback quality.

7. Reproducibility and submission

Install requirements.txt, set GROQ_API_KEY and GNANI_API_KEY in a private .env file, and keep the supplied data/ and kb/ folders available locally (or set CAPSTONE_DATASET_DIR). Run **python run.py batch <audio-folder> <output-folder> --language en-IN** for the voice path. The generated response audio is WAV. Do not publish credentials or customer/order datasets.

A short demo video has been recorded by the project author. The project folder contains the tested input/output audio artifacts; the video itself was not found in the project folder when this report was prepared. Source repository: github.com/Amogh198/CapstoneProject_VoiceOps_ZentroMart.